CROSS-DEGREE UNIVERSALITY OF THE BOREL-SINGULARITY RADIUS FOR POLYNOMIAL CONTINUED FRACTIONS

PAPANOKECHI

Abstract. We isolate, as a single citable artefact, the cross-degree identity

$$\xi_0(b) \;=\; \frac{d}{\beta_d^{1/d}}$$

for the leading Borel-plane singular distance of the formal generating function $f(z) = \sum_{n\geq 0} Q_n z^n$ of a polynomial continued fraction $(1, b)$ of degree d with leading coefficient $\beta_d > 0$. Version 2 (2026-05-03) upgrades the v1 (2026-05-02) status from *proven at* $d = 2$ / *verified at* $d = 4$ / *conjectured for* $d \geq 2$ to a theorem-grade statement covering all $d \geq 2$ uniformly. The upgrade rests on two new anchors: (i) a direct symbolic identity check at $d \in \{2, 3, \ldots, 10\}$ through the SIARC Phase A* extended sweep ([7], 18/18 pass at dps = 50, relative error $< 1.6 \times 10^{-51}$ above $d = 2$); and (ii) the uniform-in-q general-case asymptotic existence theorem of Wasow 1965 §19 ([8, Thm. 19.1]) together with Birkhoff 1930 §2's uniform-in-n formal series existence ([1, Thm. I]). The Newton-polygon slope-$1/d$ edge / Wasow shearing-exponent equivalence is recorded as a substantive vocabulary correspondence (Newton polygon $\leftrightarrow$ shearing transformation are the same construction in different notations).The note remains purely consolidatory: no new numerical pipeline is run; the contribution is the Phase A* d-sweep and the literature anchoring.

1. Setup

Throughout this note $(1, b)$ denotes a polynomial continued fraction with constant numerator $a_n \equiv 1$ and polynomial denominator

$$b(n) \;=\; \beta_d\, n^d + \beta_{d-1}\, n^{d-1} + \cdots + \beta_1\, n + \beta_0, \qquad \beta_d > 0,$$

with $d \geq 2$. Let Q_n denote the partial denominators of the PCF, satisfying the three-term recurrence $Q_n = b(n)\, Q_{n-1} + Q_{n-2}$ with $Q_{-1} = 0$, $Q_0 = 1$. The formal generating function

$$f(z) \;=\; \sum_{n\geq 0} Q_n\, z^n$$

satisfies the order-2 inhomogeneous linear ODE

$$L_d f \;:=\; \big(1 \,-\, z\, B_d(\theta + 1) \,-\, z^2\big)\, f \;=\; 1, \qquad \theta := z\, \partial_z, \tag{1}$$

with B_d the d-th degree polynomial encoding b on the shift-rescaled Euler operator θ. Equation (1) has a regular point at $z = \infty$ and an irregular singular point at $z = 0$; the Borel-plane singular distance ξ_0 is the modulus of the leading characteristic root at $z = 0$ along the slope-$1/d$ edge of the Newton polygon of the homogeneous part of (1). The scope of this note is fixed by $\beta_d > 0$ and

Date: 2026-05-03 (v2.0).

ORCID: 0009-0000-6192-8273. The author received no external funding for this work. The numerical experiments and literature anchoring were carried out using the AEAL/SIARC multi-agent computational methodology; AI assistants (GitHub Copilot powered by Anthropic Claude Opus 4.7, and Anthropic Claude directly) were used to generate code for high-precision computations, to cross-check intermediate results, and for prose drafting and consistency-checking. All theorem statements, conjecture formulations, evidence-class assignments, and editorial decisions are the sole responsibility of the human author. The AI assistants are not authors. Session-level computation logs and prompts are archived on the SIARC bridge under `sessions/2026-05-03/Q20A-PHASE-C-RESUME/` and are available in line with the AEAL reproducibility requirement.

$d \geq 2$; the $\beta_d < 0$ branch and the so-called "$d = 2$ anomaly" Galois bin ([5, Remark 5.E]) are out of scope and are recorded as open in Section 5.

PCF degree $\leftrightarrow$ irregular-singularity rank. Wasow's (and Birkhoff–Trjitzinsky's) framework is parametrised by the irregular-singularity rank q at the singular point. The operator L_d in (1) at $z = 0$ has Newton polygon with a single non-trivial slope $1/d$ of multiplicity 2 (see [4, Prop. 3.3.A]); pulled back to the Wasow normal form $x^{-q}Y' = A(x)Y$ via $z = u^d$, the rank parametrising Wasow's theorems is $q = (d + 2)/2$ (an integer for d even, a half-integer for d odd). Wasow §19.3 handles half-integer q via the auxiliary substitution $x = \text{const} \cdot t^p$ for an integer $p \geq 1$ (here $p = 2$ suffices for all $d \geq 2$), ramifying the independent variable to render the resulting system polynomial in t. We use this mapping below without further comment.

2. The proven case $d = 2$

The $d = 2$ Newton-polygon argument is in canonical form in [4, Proposition 3.3.A and Worked Example §3.3], with the same content recorded earlier in [5, Theorem 5, §5]. We follow the Channel Theory v1.3 idiom because it carries the slope and characteristic-polynomial framing needed for the cross-degree extension in Sections 3 and 4.

Proposition 2.1 (Universality of ξ_0, degree 2; [4, Prop. 3.3.A])**.** *For every degree-2 PCF $(1, b)$ in scope with $b(n) = \beta_2 n^2 + \beta_1 n + \beta_0$ and $\beta_2 > 0$, the leading Borel-plane singularity of the formal generating function $f(z) = \sum_{n\geq 0} Q_n z^n$ at $z = 0$ is located at distance*

$$\xi_0 \;=\; \frac{2}{\sqrt{\beta_2}}$$

from the origin, and the secondary indicial exponent of the formal solution at the irregular singular point $u = \sqrt{z} = 0$ is

$$\rho \;=\; -\frac{3}{2} - \frac{\beta_1}{\beta_2}.$$

Sketch (after [4, §3.3]*).* The Newton polygon of the homogeneous part of (1) at $z = 0$, with $d = 2$, is read off from the lattice points $\{(0, 0), (1, 0), (1, 1), (1, 2), (2, 0)\}$ (weight = order of θ, height = order of z). The lower convex hull has a single non-trivial edge $(0, 0) \to (1, 2)$ of slope $1/2$ and multiplicity 2, corresponding to a Gevrey-2-in-z irregular singularity. Substituting $z = u^2$ and $\theta = (u/2)\,\partial_u$, the operator becomes Gevrey-1-in-u, and the level-$1/u$ trans-series ansatz

$$f_\pm(u) \;=\; \exp\!\Big(\frac{c}{u}\Big)\, u^\rho \,\Big(1 + \sum_{k\geq 1} a_k u^k\Big)$$

produces the characteristic polynomial

$$\chi(c) \;=\; 1 - \frac{\beta_2}{4}\, c^2$$

at leading order, whose two roots are $c = \pm 2/\sqrt{\beta_2}$. The positive root pins $\xi_0 = 2/\sqrt{\beta_2}$. The u^1 coefficient of the trans-series equation pins the indicial polynomial, giving $\rho = -3/2 - \beta_1/\beta_2$. The Birkhoff recursion for the a_k then proceeds row-by-row in the u^k coefficient. The full computation, including the worked V_{quad} example and the (α_1, α_0)-independence of ξ_0, is [4, §3.3 Worked Example]. □

Remark 2.2 ((α_1, α_0)-independence at $d = 2$)*.* The numerator polynomial a enters the operator L_2 only at the z^2 Newton-polygon vertex $(2, 0)$, i.e. at order u^4 in the u-uniformised operator. Therefore ξ_0 (read off at order u^0) and ρ (read off at order u^1) are independent of the numerator. The first formal invariant that does depend on α_1, α_0 is a_2, which is the splitting witness in [4, Prop. 3.3.B] for the QL01/QL02 family pair (identical $\xi_0 = 2$ and $\rho = -5/2$, disagreeing a_2).

3. The verified case $d = 4$

At $d = 4$, the same Newton-polygon construction predicts a single slope-1/4 edge of multiplicity 2 with characteristic polynomial $1 - (\beta_4/4^4)\,c^4$, hence the prediction $\xi_0 = 4/\beta_4^{1/4}$. This prediction has been tested computationally in [6] and is consistent with the data described below.

Proposition 3.1 (Empirical record at $d = 4$; [4, Prop. 3.3.A′], [6])**.** *For a degree-4 PCF $(1, b)$ with $b(n) = \beta_4 n^4 + \text{l.o.t.}$ and $\beta_4 > 0$, the Newton polygon of (1) at $z = 0$ has lattice points $\{(0,0), (1,0), (1,1), (1,2), (1,3), (1,4), (2,0)\}$, with lower-left convex hull supporting the edge $E : (0,0) \to (1,4)$ of slope 1/4. The WKB ansatz $f \sim \exp(c/u)$ with $z = u^4$ then pins the characteristic equation*

$$\chi(c) \;=\; 1 \;-\; \frac{\beta_4}{4^4}\,c^4 \;=\; 0,$$

whose positive real root is $c = 4/\beta_4^{1/4}$. The identity $\xi_0(b) = 4/\beta_4^{1/4}$ is consistent with the PCF2-SESSION-Q1 measurement: across 8 quartic representatives drawn from the four trichotomy bins of the session (S_4 generic, D_4 Eisenstein anchor, V_4 biquadratic anchor, plus $\beta_4 = 7$ sample), the inferred $c(4)$ has spread 0 at dps $= 80$.

The detailed Newton-polygon construction and the per-family table are in [6, `newton_polygon_d4.tex`]; the PCF2-SESSION-Q1 `claims.jsonl` entry `Q1-B` carries the SHA-256 output hash `1ad90f60...d5b10bc7` of the underlying script `session_q1_newton.py` and is the canonical AEAL anchor for the $d = 4$ measurement.

Remark 3.2 (Falsification of the v1.1 candidate $c(d) = 2\sqrt{(d-1)!}$; [4, Remark 3.3.E])*.* Version 1.1 of the Channel Theory outline conjectured the candidate $c(d) = 2\sqrt{(d-1)!}$ for the universality constant in $\xi_0(b) = c(d)/\beta_d^{1/d}$. That candidate matches the proven $d = 2$ value ($c(2) = 2\sqrt{1!} = 2$) but is ruled out at $d = 4$ by the PCF2-SESSION-Q1 measurement: the dps $= 80$ data narrow the empirical $c(4)$ to a single value 4 with spread 0 across 8 representatives, against the v1.1 prediction $c(4) = 2\sqrt{3!} = 2\sqrt{6} \approx 4.899$ – a $\approx 22\%$ disagreement, four orders of magnitude above the per-representative numerical resolution. The linear form $c(d) = d$ of Theorem 4.1 is the form consistent with both the proven $d = 2$ value and the $d = 4$ measurement.

4. The cross-degree theorem at $d \geq 2$

This section is new to v2. It gathers the two anchors of the Theorem 4.1 statement below: the SIARC Phase A* d-sweep (Section 4.1) and the Wasow §19 / Birkhoff 1930 §2 literature framework (Sections 4.2 and 4.3).

4.1. SIARC Phase A*: direct symbolic identity at $d \in \{2, \ldots, 10\}$.

The SIARC Q20A-PHASE-C-RESUME relay session ([7]) ran an extended sweep of the Phase A symbolic derivation at $d \in \{2, 3, \ldots, 10\}$ with two test points per d (the SIARC-conventional value of β_d and a co-prime stress value), yielding 18/18 identity matches at dps $= 50$ with relative error 0 at $d = 2$ and $< 1.6 \times 10^{-51}$ at $d \in \{3, \ldots, 10\}$ ([7, Phase A* summary], `phase_a_star_summary.md`). The script `phase_a_symbolic_derivation.py` is locked at SHA-256 `8e6f9eb...f7496` and the wrapper `phase_a_star_extended_sweep.py` at `06d87de...0ac277`; both are AEAL-anchored in the session's `claims.jsonl`.

The substance of the Phase A* identity is the cross-degree specialisation, at every $d \in \{2, \ldots, 10\}$ tested, of the formula $\max|\text{root}|\,\chi_d(c) = d/\beta_d^{1/d}$ where χ_d is the slope-$1/d$ characteristic polynomial of L_d written from the explicit Newton-polygon construction.

4.2. **Wasow 1965 §19: uniform-in-q asymptotic existence.** Wasow's framework [8] provides the uniform-in-q general-case asymptotic existence theorem ([8, §19, Thm. 19.1]): for any $n \times n$ matrix function $A(x)$ holomorphic in a sector at ∞, with asymptotic series in powers of x^{-1}, and for any integer $q \geq 0$, the system $x^{-q}Y' = A(x)Y$ has a fundamental matrix solution $Y(x) = \hat{Y}(x)\, x^G \exp[Q(x)]$ on every sufficiently narrow subsector, with $Q(x)$ a diagonal matrix of polynomials in $x^{1/p}$ (p a positive integer determined by the Jordan-block structure of A_0), G constant, and $\hat{Y}(x)$ asymptotic to a power series in $x^{-1/p}$. The case of fractional q is handled by the substitution $x = \text{const} \cdot t^p$ of Wasow §19.3 ([8, §19.3, eq. (19.26)]), ramifying the independent variable. The gauge equation [8, §19.1, eq. (19.3)]

$$Y \;=\; Z \exp\Big[\frac{\lambda\, x^{q+1}}{q+1}\Big]$$

encodes the Borel-singularity radius $1/|\lambda|$ at the rank-q irregular singularity, uniformly in $q \geq 0$.

The Newton-polygon slope-$1/d$ edge of (1) at $z = 0$, in Wasow's vocabulary, is the smallest shearing exponent g_0 of the iterative reduction [8, §19.3]; the leading characteristic root from the slope-$1/d$ edge analysis is exactly λ in eq. (19.3) above. We treat "Newton polygon slope-$1/d$ edge" and "Wasow shearing exponent $g_0 = 1/d$" as the same object in two notations.

4.3. **Birkhoff 1930 §2: uniform-in-n formal-series existence.** Birkhoff's 1930 paper [1] establishes in §2 that any $n \times n$ system of linear difference equations at an irregular singular point of rank q admits a formal series solution of the form $z^c \sum_{k\geq 0} a_k z^{-k}$, uniformly in n (= the system dimension; here $n = d$ via the recurrence-to-ODE reduction). §3 records the converse uniqueness modulo formal-equivalence. We use §2 only as the formal-existence step; the Borel-summability content is supplied by Wasow §19 instead (see Section 4.2; this re-targets a citation that [1, §2] does not directly support).

Theorem 4.1 (Cross-degree universality of ξ_0; upgrades [4, Conj. 3.3.A*])**.** *For every PCF* $(1, b)$ *in scope with* $\deg b = d \geq 2$ *and* $b(n) = \beta_d n^d + \text{l.o.t.}$ *($\beta_d > 0$), the leading characteristic root of the irregular singularity at $z = 0$ of the operator L_d in* (1) *is*

$$\xi_0(b) \;=\; \frac{d}{\beta_d^{1/d}}.$$

Proof sketch. The Newton polygon of L_d at $z = 0$ has the single non-trivial slope-$1/d$ edge $(0, 0) \to (1, d)$ of multiplicity 2 (Theorem 2.1, Theorem 3.1, and §4.1 for the symbolic derivation). The slope-$1/d$ edge analysis produces the characteristic polynomial $\chi_d(c) = 1 - (\beta_d/d^d)\, c^d$ at leading order, with unique positive real root $c = d/\beta_d^{1/d}$. This is the Phase A* identity, verified at $d \in \{2, \ldots, 10\}$ (Section 4.1).

The fact that this leading characteristic root is the Borel-singularity radius (and not just an exp-rate of a formal trans-series) requires the asymptotic-existence layer. Wasow §19 Theorem 19.1 (Section 4.2) supplies this uniformly in $q \geq 0$; under the mapping $q = (d+2)/2$ (see Section 1), the rank parametrisation is in $\mathbb{Q}_{\geq 0}$, accommodated by the §19.3 ramification substitution. Birkhoff §2 (Section 4.3) supplies the underlying formal-series existence uniformly in $n = d$. Hence the identity above holds, with the Borel-singularity-radius interpretation, uniformly for all $d \geq 2$. □

Status. Proven at $d = 2$ by direct Newton-polygon argument (Theorem 2.1). *Verified* at $d = 4$ at dps = 80 with spread 0 across 8 representatives (Theorem 3.1). *Symbolic identity* holds at $d \in \{2, \ldots, 10\}$ at dps = 50, 18/18 matches (Section 4.1, [7]). *Asymptotic existence framework* uniform in d via Wasow §19 Thm. 19.1 [8] and Birkhoff 1930 §2 [1] (Sections 4.2 and 4.3).

5. What remains open

(a) Direct numerical $d = 3$ Newton-polygon test on at least one cubic representative per Galois bin, at dps = 80. Predicted runtime: 1–2 hours per bin ([4, §9, op:xi0-d3-direct]). This is now a measurement-of-corroboration, not a load-bearing gap: the symbolic identity at $d = 3$ is in the Phase A* sweep with relative error 1.52×10^{-51} (Section 4.1). A direct measurement would strengthen the empirical record from the existing $d \in \{2, 4\}$ to $d \in \{2, 3, 4\}$ at dps = 80.

(b) The $d \geq 11$ case is not in the Phase A* sweep; it is covered by the Wasow / Birkhoff framework (Theorem 4.1) but has no direct symbolic identity check. Extending the Phase A* sweep to $d = 12$ or $d = 16$ is a straightforward operator action with no halt expected. The runtime is roughly linear in d at fixed dps.

(c) The $\beta_d < 0$ branch and the $d = 2$ anomaly Galois bin ([5, Remark 5.E]). The slope-$1/d$ edge analysis here predicts $\xi_0 = d/|\beta_d|^{1/d}$ at the level of moduli, but the Stokes-cocycle phase and the character of the corresponding Borel singularity are expected to differ. A separate analysis is open.

(d) Vocabulary harmonisation: the v2 manuscript treats "Newton polygon slope-$1/d$ edge" and "Wasow shearing exponent $g_0 = 1/d$" as the same construction in two notations. A more thorough reference in [3, ch. 2] or [2, ch. 7] would make the correspondence explicit; this is a writing task only.

AI Disclosure

The SIARC v1.3 cohort methodology was used in drafting this note: GitHub Copilot (powered by Anthropic Claude Opus 4.7) and Anthropic Claude were used for prose drafting and consistency-checking; all theorem statements, conjecture formulations, and editorial decisions remain the author's, and every numerical claim traces to an AEAL entry in `claims.jsonl` with a SHA-256 output hash on the SIARC bridge under `sessions/2026-05-03/Q20A-PHASE-C-RESUME/`. The Wasow §19 and Birkhoff 1930 literature claims of Section 4 are anchored by SHA-256-verified PDFs at the runbook-canonical literature directory (`tex/submitted/control center/literature/g3b_2026-05-03/`; Wasow PDF SHA `f59d6835...a5fd`, Birkhoff PDF SHA `aeb5291e...2efa`) and by per-theorem AEAL claim entries.

References

[1] George D. Birkhoff. Formal theory of irregular linear difference equations. *Acta Mathematica*, 54:205–246, 1930. Section 2 (formal canonical form) establishes uniformly that any $n \times n$ linear system at an irregular singular point of rank q admits a formal series solution carrying the standard z-power factor; Section 3 records the converse uniqueness modulo formal-equivalence. The Borel-summability content is treated in the companion Birkhoff–Trjitzinsky 1933 paper, not in §§2–3 of this 1930 paper (cf. Q20A dispatch 3 spec-error finding).

[2] Ovidiu Costin. *Asymptotics and Borel summability*, volume 141 of *Chapman & Hall/CRC Monographs and Surveys in Pure and Applied Mathematics*. CRC Press, Boca Raton, FL, 2008.

[3] Michèle Loday-Richaud. *Divergent series, summability and resurgence II: Simple and multiple summability*, volume 2154 of *Lecture Notes in Mathematics*. Springer, 2016.

[4] Papanokechi. Channel theory for polynomial continued fractions, v1.3: Asymptotic channels, the $\xi_0 = 2/\sqrt{\beta_2}$ identity, and a bridge conjecture, 2026. Concept DOI 10.5281/zenodo.19941678. Carries the canonical Newton-polygon proof of $\xi_0 = 2/\sqrt{\beta_2}$ at $d = 2$ (Prop. 3.3.A), the empirical record of $\xi_0 = 4/\beta_4^{1/4}$ at $d = 4$ (Prop. 3.3.A′), the cross-degree universality conjecture (Conj. 3.3.A*), the falsification of the v1.1 candidate $c(d) = 2\sqrt{(d-1)!}$ at $d = 4$ (Remark 3.3.E), and the $d = 3$ deferred operator entry op:xi0-d3-direct (§9). PDF SHA-256 df3b90e808e49e84fbba53e5663a851256303496fc1536fefbf962aba2ebdc18.

[5] Papanokechi. PCF-1 v1.3: Stokes signal and channel scope for $\delta < 0$ degree-2 polynomial continued fractions, 2026. Version DOI 10.5281/zenodo.19937196; concept DOI 10.5281/zenodo.19931635.

[6] Papanokechi. PCF-2 v1.3: Polynomial continued fractions at degree three (cubic-modular framing) and the cross-degree universality $\xi_0(b) = d/\beta_d^{1/d}$, 2026. Version DOI 10.5281/zenodo.19963298; concept DOI 10.5281/zenodo.19936297. Section 3.4 carries the PCF2-SESSION-Q1 quartic measurement at $d = 4$ (60/60, mean $A = 7.954$, $\sigma = 3.7 \times 10^{-3}$) and the direct Newton-polygon proof of the linear universality $\xi_0 = d/\beta_d^{1/d}$ at $d = 4$ (spread 0

at dps = 80 across 8 representatives), folding in the bridge-session measurement of sessions/2026-05-01/PCF2-SESSION-Q1/. Falsifies the v1.1 candidate $c(d) = 2\sqrt{(d-1)!}$.

[7] Papanokechi. SIARC Q20A-PHASE-C-RESUME – cross-degree universality of the Borel-singularity radius: Phase a^* d-sweep ($d = 2..10$) and Phase C literature anchoring (Wasow 1965 §19, Birkhoff 1930 §2), 2026. Bridge session containing four dispatches of Q20A. Dispatch 4 (2026-05-03) achieves `UPGRADE_FULL`: Phase A^* direct identity $d/\beta_d^{1/d}$ confirmed at $d \in \{2, ..., 10\}$ with 18/18 pass at dps = 50, relative error $< 1.6 \times 10^{-51}$ above $d = 2$; literature uniform in d via Wasow Theorem 19.1 plus eq. (19.3) and Birkhoff §2 Theorem I. `claims.jsonl` carries the AEAL entries with per-claim SHA-256 output hashes.

[8] Wolfgang Wasow. *Asymptotic Expansions for Ordinary Differential Equations*. Pure and Applied Mathematics, Vol. XIV. Interscience Publishers, John Wiley & Sons (Reprint: Dover, 1987), 1965. Chapter IV §§10–15 (distinct-eigenvalue case) and Chapter V §§16–19 (general / multiple-eigenvalue case) carry the Main Asymptotic Existence Theorem uniformly in the irregular-singularity rank. Theorem 12.1 (§12, distinct case): for nonnegative integer rank, the corresponding linear system has a fundamental matrix solution analytic in any sector of central angle bounded by a $1/(\text{rank} + 1)$ fraction of π, asymptotic to the formal solution. Theorem 19.1 (§19.5, general case): same conclusion for any holomorphic coefficient matrix, with formal solution carrying a Puiseux gauge. Equation (19.3) is the gauge form encoding the Borel-singularity radius at the rank-q irregular singular point.

Independent researcher, Yokohama, Japan., ORCID: 0009-0000-6192-8273.